

Infrared Sensor Based POWER SAVER

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Quite often you forget to switch off the light or fan when going out of the room. The simple circuit presented here will automatically switch off electrical appliances like lights or fans as you vacate a room, after a predetermined time period. It will also switch on the light when you enter the room again. This will reduce unnecessary power consumption. Fig. 1 shows the pyroelectric/passive infrared (PIR) motion sensor used in the circuit.

Circuit and working

The circuit diagram of the PIR sensor based power saver is shown in Fig. 2. It is built around bridge rectifier DB107 (BR1), PIR motion sensor connected across connector CON2, timer NE555 (IC1), two 1N4007 rectifier diodes (D1 and D2) and a few other components.

The circuit uses a PIR sensor, which detects the presence of people through change in the infrared radiation from the room when

people enter or leave the room. The PIR sensor outputs around 3.3V high signal whenever it detects radiation change in front of it.

IC1, resistor R3, potmeter VR1 and capacitor C3 are used as a timer here to convert small time span of PIR signal to a long delay. Output of IC1 at pin 3 drives transistor T2, which, in turn, controls relay RL1. Electrical loads like lights or fans are controlled through this relay.

230V AC mains power is connected across connector CON1. It is stepped down to 9V through transformer X1, rectified by bridge rectifier BR1 and filtered by capacitor C1. Thus we get around 9V DC at test point TP1. This 9V DC voltage is used as power supply for the circuit.

When the circuit is first switched on, capacitor C3 charges through potmeter VR1 and resistor R3. During this time, voltage at pins 2 and 6 of IC1 is less than two-thirds of its supply voltage, and so output pin 3 goes high. This energises the relay through

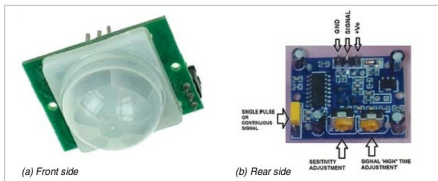


Fig. 1: PIR motion-sensor module

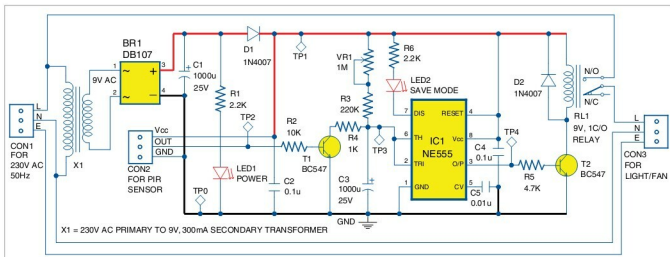


Fig. 2: Circuit diagram of the PIR sensor based power saver